E-commerce System Architecture and Flows

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About

This diagram illustrates the main functional and application flows of an e-commerce ecosystem, from product design and production planning to customer order and delivery.
The architecture is structured into two main layers: Back End and Front End, distinguishing internal operational systems from customer interaction channels.

When I first entered the retail sector, I wished I had a diagram showing the end-to-end systems interacting within an Information System.
That is why I decided to create this diagram: to provide a clear, global and simplified view of the landscape.
My objective was to clarify the many acronyms and concepts used in retail IT by bringing them together in one place, making the ecosystem easier to understand.

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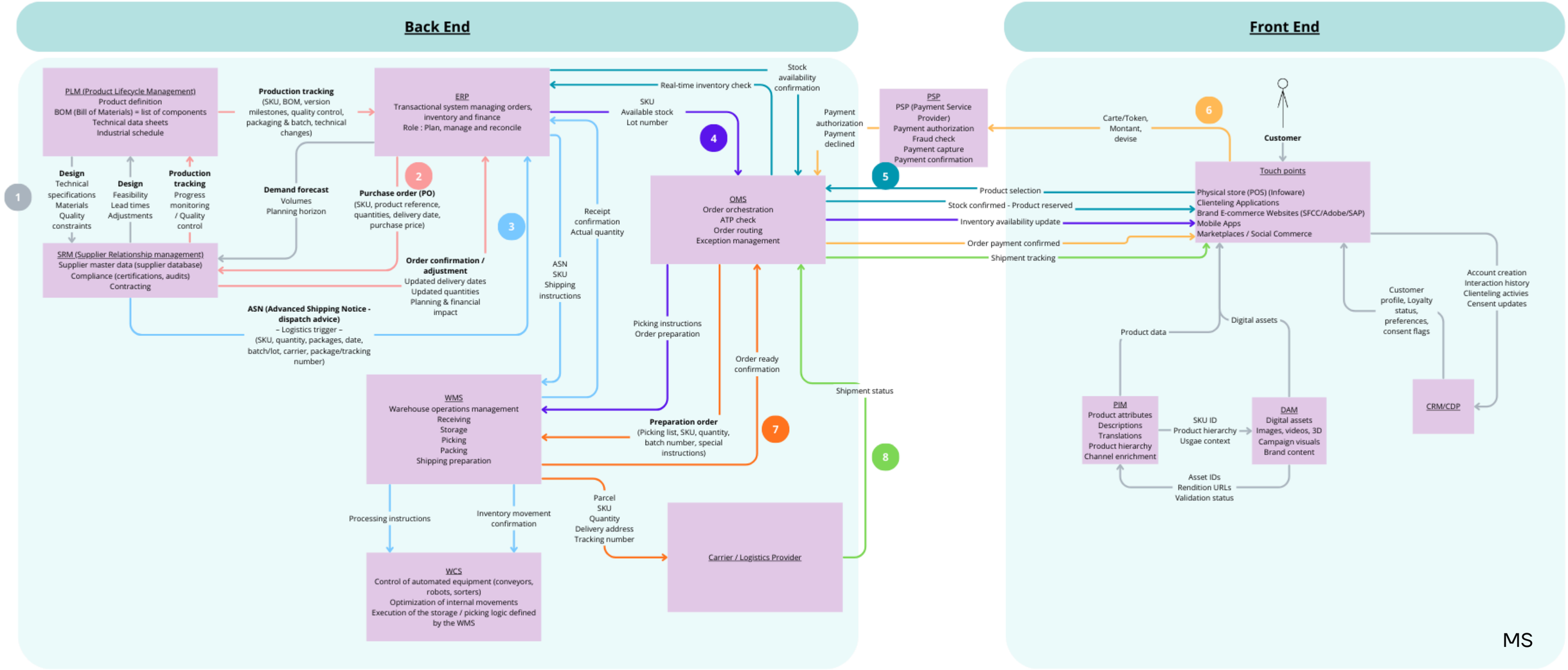
Architecture overview



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Legend

1

Product design and planning

The cycle begins with PLM and SRM systems, where product information is defined (technical sheets, BOM, quality constraints) as well as supplier relationships.
These elements feed demand forecasting processes and enable the creation of Purchase Orders (PO) in the ERP system.

2

Order and production management

The ERP centralizes transactional processes such as supplier orders, production planning, and financial reconciliation.
Order confirmations and adjustments are integrated to ensure consistency between procurement, production planning, and financial processes.

3

Logistics and warehouse operations

Logistics flows are triggered through ASN (Advanced Shipping Notices) sent by suppliers.
The WMS then manages warehouse operations including receiving, storage, picking, packing, and shipment preparation.
The WCS controls automated warehouse equipment such as conveyors, robots, and sorting systems.

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Legend

4

Order orchestration

The OMS plays a central role in the e-commerce architecture.
It orchestrates customer orders, performs real-time stock availability checks, determines the optimal fulfillment location, and manages routing and exceptions.

5

Order and production management

Payments are handled through a PSP (Payment Service Provider), which manages authorization, fraud detection, and transaction confirmation.

7

8

Shipping and delivery tracking

Once the order is prepared, parcels are transferred to the carrier or logistics provider.
Shipping status and tracking information are then sent back to the OMS and front-end interfaces to provide customers with real-time delivery visibility.

6

Customer experience and digital channels

Customers interact with the ecosystem through multiple touchpoints, including e-commerce websites, mobile applications, physical stores (POS), and marketplaces.
These channels rely on product data from the PIM, digital assets from the DAM, and customer data from the CRM/CDP.

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